

Amin Saberi

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EDUCATION

- 2022–26 Ph.D., Heinrich Heine University Düsseldorf, Germany *summa cum laude*
Thesis: "Computational Analysis of Multiscale Cortical Organization and Development"
- 2011–19 M.D., Mashhad University of Medical Sciences, Iran

PROFESSIONAL EXPERIENCE

- 2025– Max Planck Institute for Human Cognitive and Brain Sciences, Germany
Postdoctoral Researcher
- 2025– Research Center Jülich, Germany
Guest Researcher
- 2021–25 Max Planck Institute for Human Cognitive and Brain Sciences / Research Center Jülich, Germany
Doctoral Researcher
- 2018–21 Institute of Medical Science and Technology, Iran
Remote Guest Researcher
- 2017–18 Mashhad University of Medical Sciences, Iran
Medical Intern

RESEARCH AREAS

Brain imaging, Computational neuroscience, Psychiatry, Meta-analyses, Brain dynamics, Excitation-inhibition balance, Neurodevelopment

SKILLS

Programming (Python, C/C++, CUDA, R & Matlab), Functional and structural MRI analysis, Brain network modelling, Coordinate-based and effect-size-based meta-analyses, Statistical analysis, High performance computing, Git/GitHub, Datalad

PUBLICATIONS

[Link to Google Scholar](#)

First and Last/Corresponding Authorship

- 2026 Saberi, A.*, Wan, B., Wischnewski, K. J., Jung, K., Sasse, L., Hoffstaedter, F., ... Valk, S. L.* **"cuBNM: GPU-Accelerated Brain Network Modeling."** *eLife*. doi:10.7554/eLife.111817
- 2025 Magielse, N.*, Manoli, A., Eickhoff, S. B., Fox, P. T., Saberi, A.[†], Valk, S. L.[†]* **"A bias-accounting meta-analytic approach refines and expands the cerebellar behavioral topography."** *Neuroscience & Biobehavioral Reviews*, 179, 106418. doi:10.1016/j.neubiorev.2025.106418
- 2025 Saberi, A., Wischnewski, K. J., Jung, K., Lotter, L. D., Schaare, H. L., Banaschewski, T., ... Valk, S. L.* **"Adolescent maturation of cortical excitation-inhibition ratio based on individualized biophysical network modeling."** *Science Advances*, 11(23), eadr8164. doi:10.1126/sciadv.adr8164

- 2024 Saberi, A., Ebneabbasi, A., Rahimi, S., Sarebannejad, S., Sen, Z. D., Graf, H., ... Tahmasian, M.* **“Convergent functional effects of antidepressants in major depressive disorder: A neuroimaging meta-analysis.”** *Molecular Psychiatry*. doi:10.1038/s41380-024-02780-6
- 2023 Saberi, A., Paquola, C., Wagstyl, K., Hettwer, M., Bernhardt, B.C., Eickhoff, S.B., Valk, S.L.* **“The regional variation of laminar thickness in the human isocortex is related to cortical hierarchy and interregional connectivity.”** *PLOS Biology*. doi:10.1371/journal.pbio.3002365
- 2021 Saberi, A., Mohammadi, E., Zarei, M., Eickhoff, S. B., Tahmasian, M.* **“Structural and functional neuroimaging of late-life depression: A coordinate-based meta-analysis.”** *Brain Imaging and Behavior*, 16, 518–531. doi:10.1007/s11682-021-00494-9
- 2021 Saberi, A., Abdolalizadeh, A., Mohammadi, E., Nahayati, M. A., Bagheri, H., Shekarchi, B., Kargar, J.* **“Thalamic shape abnormalities in patients with multiple sclerosis-related fatigue.”** *Neuroreport*, 32(6), 438–442. doi:10.1097/WNR.0000000000001616
- 2021 Bijari, M., Jangjoo, S., Emami, N., Raji, S., Mottaghi, M., Moallem, R., Jangjoo, A., Saberi, A.* **“The Accuracy of Visceral Adiposity Index for the Screening of Metabolic Syndrome: A Systematic Review and Meta-Analysis.”** *International Journal of Endocrinology*, 2021, e6684627. doi:10.1155/2021/6684627
- 2017 Ganji, A., Esmacilzadeh, A., Ghanaei, O., Saberi, A.*, Taherzadeh, D., Sazgarnia, S., Mayabi Joghali, Z., Zirak, M., AbdollahRamazani, S., Zarifmahmoudi, L. **“Predictive value of red blood cell distribution width for mortality in patients with acute pancreatitis: A systematic review and meta-analysis.”** *Medical Journal of the Islamic Republic of Iran*, 31, 124. doi:10.14196/mjiri.31.124

Co-authorship

- 2026 Afshani, M., Saberi, A., Reimann, G. M., Chu, C., Elmenhorst, D., ..., Tahmasian, M.* **“Convergent connectivity of sleep disorders and its neurotransmitter and cell enrichment correlates.”** *medRxiv*. doi:10.64898/2026.09.09.26362596
- 2026 Hettwer, M. D.*, Saberi, A., Shafiei, G., Manoli, A., Boer, A. A. D., Alnæs, D., ... Valk, S. L.* **“Lifespan brain structural variation reveals shared organization across mental health conditions.”** *medRxiv*. doi:10.64898/2026.08.13.26360304
- 2026 John, A.*, Saberi, A., Manoli, A., Royer, J., Erigüec, D. Y., Sydnor, V. J., Wan, B., Eickhoff, S. B., Bernhardt, B. C., Anwender, A., Valk, S. L.* **“Nucleus-level thalamic organization anchors multimodal signatures of thalamocortical maturation.”** *bioRxiv*. doi:10.64898/2026.07.03.736332
- 2026 Ferrari, A., Wan, B., Kabbeck, J., Saberi, A., Kaiser, S., ... Kirschner, M.* **“Categorical and Dimensional Alterations Along Two Principal Cortical Gradient Axes Across the Schizophrenia-Bipolar Spectrum.”** *medRxiv*. doi:10.64898/2026.06.30.26356921
- 2026 Bi, H., Åkerstedt, T., Bülow, R., ... Saberi, A., ... Tahmasian, M.*, ENIGMA-Sleep Working Group. **“Prediction of cognitive performance by demographics, sleep, and brain morphometry: Machine learning findings from ENIGMA-Sleep Working Group.”** *Research Square*. doi:10.21203/rs.3.rs-9914920/v1
- 2025 John, A. *, Hettwer, M. D., Schaare, H. L., Saberi, A., Bayrak, Ş., Wan, B., ... Valk, S. L.* **“A multimodal characterization of low-dimensional thalamocortical structural connectivity patterns.”** *Communications Biology*, 8(1), 1–15. doi:10.1038/s42003-025-07528-8
- 2024 Wan, B. *, Saberi, A., Paquola, C., Schaare, H. L., Hettwer, M. D., Royer, J., ... Valk, S. L.* **“Microstructural asymmetry in the human cortex.”** *Nature Communications*, 15(1), 10124. doi:10.1038/s41467-024-54243-9
- 2024 Lotter, L. D., Saberi, A., Hansen, J. Y., Misic, B., Paquola, C., Barker, G. J., ... Dukart, J.* **“Regional patterns of human cortex development correlate with underlying neurobiology.”** *Nature Communications*, 15(1), 7987. doi:10.1038/s41467-024-52366-7

- 2024 Küchenhoff, S., Bayrak, Ş., Zsido, R. G., Saberi, A., Bernhardt, B. C., Weis, S., ... Valk, S. L.* **“Relating sex-bias in human cortical and hippocampal microstructure to sex hormones.”** *Nature Communications*, 15(1), 1–19. doi:10.1038/s41467-024-51459-7
- 2024 Elberse, J. D., Saberi, A., Ahmadi, R., Changizi, M., Bi, H., Hoffstaedter, F., ... Tahmasian, M.* **“The interplay between insomnia symptoms and Alzheimer’s Disease across three main brain networks.”** *Sleep*, zsa145. doi:10.1093/sleep/zsae145
- 2023 Maleki Balajoo, S., Eickhoff, S. B., Masouleh, S. K., Plachti, A., Waite, L., Saberi, A., Bahri, M. A., Bastin, C., Salmon, E., Hoffstaedter, F., Palomero-Gallagher, N., Genon, S.* **“Hippocampal metabolic subregions and networks: Behavioral, molecular, and pathological aging profiles.”** *Alzheimer’s & Dementia*. doi:10.1002/alz.13056
- 2022 Hettwer, M. D., Saberi, A., Fan, Y.-S., Valk, S. L.* **“Schizophrenia and Macroscale Brain Structure: Genes in Context.”** *Biological Psychiatry*, 92(4), 258–260. doi:10.1016/j.biopsych.2022.06.010
- 2022 Kamalian, A.[†], Khodadadifar, T.[†], Saberi, A., Masoudi, M., Camilleri, J. A., Eickhoff, C. R., ... Tahmasian, M.* **“Convergent regional brain abnormalities in behavioral variant frontotemporal dementia: A neuroimaging meta-analysis of 73 studies.”** *Alzheimer’s & Dementia: Diagnosis, Assessment & Disease Monitoring*, 14(1), e12318. doi:10.1002/dad2.12318
- 2022 Ghodsi, H., Rahimi, H. R., Aghili, S. M., Saberi, A., Shoeibi, A.* **“Evaluation of curcumin as add-on therapy in patients with Parkinson’s disease: A pilot randomized, triple-blind, placebo-controlled trial.”** *Clinical Neurology and Neurosurgery*, 107300. doi:10.1016/j.clineuro.2022.107300
- 2022 Rahimi Jafari, S.[†], Sareban Nejad, S.[†], Saberi, A., Khazaie, H., Camilleri, J. A., Eickhoff, C. R., Eickhoff, S. B., Tahmasian, M.* **“Is there any consistent structural and functional brain abnormality in narcolepsy? A meta-analytic perspective.”** *Neuroscience & Biobehavioral Reviews*, 132, 1181–1182. doi:10.1016/j.neubiorev.2021.10.034
- 2021 Tahmasian, M.*, Aleman, A., Andreassen, O. A., Arab, Z., Baillet, M., Benedetti, F., Bresser, T., Bright, J., Chee, M. W. L., Chylinski, D., Cheng, W., Deantoni, M., Dresler, M., Eickhoff, S. B., Eickhoff, C. R., Elvsåshagen, T., Feng, J., Foster-Dingley, J. C., Ganjgahi, H., ... Saberi, A., ... Zarei, M. **“ENIGMA-Sleep: Challenges, opportunities, and the road map.”** *Journal of Sleep Research*, e13347. doi:10.1111/jsr.13347
- 2021 Ghodsi, A., Bijari, M., Alamdaran, S. A., Saberi, A., Mahmoudabadi, E., Balali, M. R., Ghahremani, S.* **“Chest computed tomography findings of COVID-19 in children younger than 1 year: A systematic review.”** *World Journal of Pediatrics*, 17, 234–241. doi:10.1007/s12519-021-00424-1
- 2021 Khorrampazhouh, N., Omranzadeh, A., Fazeli, B., Zarifian, A., Ghodsi, A., Amirkhanlou, F., Saberi, A., ... Moghadam, M. D.* **“A Systematic Review and Meta-analysis of Clinical Studies on Ankylosing Spondylitis and Neutrophil to Lymphocyte Ratio.”** *Current Rheumatology Reviews*, 18, 160–167. doi:10.2174/1573397117666210921114431
- 2020 Rezaeitalab, F., Esmaeili, M.*, Saberi, A., Vahidi, Z., Emadzadeh, M., Rahimi, H. R., Ramezani, N., Mirshabani-Toloti, S. Z. **“Predictive value of inflammatory markers for functional outcomes in patients with ischemic stroke.”** *Current Journal of Neurology*, 19(2), 47–52. doi:10.18502/cjn.v19i2.4940

* corresponding author, [†] equal contribution

TALKS & SYMPOSIA

- 2026 Symposium talk, *OHBM SEA-SIG Webinar* (Online)
- 2026 Invited talk, *Center for Neuroscience and Cognitive Systems* (Rovereto, Italy)

2025	Invited talk, <i>4D Group, University of Cambridge</i> (Online)
2025	Symposium talk, <i>Computational Psychiatry Conference</i> (Tübingen, Germany)
2024	Invited talk, <i>Gradients of Brain Organization Workshop</i> (Seoul, South Korea)
2024	Contributed talk, <i>8th BigBrain Workshop</i> (Padua, Italy)
2023	Contributed talk, <i>qMRI Conference</i> (Online)
2023	Contributed talk, <i>7th BigBrain Workshop</i> (Reykjavík, Iceland)
2022	Contributed talk, <i>6th BigBrain Workshop</i> (Zadar, Croatia)
2020	Contributed talk, <i>7th Iranian Human Brain Mapping Congress</i> (Online)
2019	Contributed talk, <i>6th Iranian Human Brain Mapping Congress</i> (Tehran, Iran)

SELECTED SOFTWARE

[Link to GitHub](#) 

“cuBNM”, a toolbox for brain network modeling on GPUs.

“NiMARE-GPU”, GPU-accelerated activation likelihood estimation meta-analysis.

SUPERVISION

2025-now	Co-leading the “Systems Neuroscience” subgroup at the CNG lab <i>Weekly meetings to discuss progress and troubleshoot problems with students from BSc to PhD level</i>
2026-now	Zheng Long Lee (Max Planck School of Cognition PhD rotation, co-supervised)
2026-now	Caroline Jantzen (PhD student, co-supervised)
2026-now	JiHoon Kim (PhD student, co-supervised)
2026	Nghi Nguyen (Max Planck School of Cognition PhD rotation, co-supervised)
2025	Klara Hamker (BSc internship, co-supervised)
2024	Delaram Roozbehani (BSc internship, co-supervised)
2023	Anna-Lea Beyer (BSc internship, co-supervised)
2022	Marie Rademacher (MSc internship, co-supervised)
2022	Julia Rakowski (BSc internship, co-supervised)

TEACHING

2024	“Methods in Neuroscience” (1 session, co-taught), Leipzig University, Leipzig, Germany
2021	“Python for Data Science” (10 sessions), Online
2020	“NeuromatchAcademy, an international computational neuroscience crash course” (15 days, teaching assistant), Online

GRANTS AND AWARDS

2024	Heine Research Academies Travel Grant (1500 €) for conference in Seoul, South Korea
2023	DAAD Travel Grant (1780 €) for conference in Montreal, Canada; Travel not undertaken due to visa delay.
2023	Compute Project (grant no. eidev) at Jülich Supercomputing Center, Germany
2015	Awarded membership of National Elites Foundation, Iran

PEER REVIEW

Reviewed academic journal articles for *Nature Reviews Neuroscience* (as co-reviewer), *Nature Human Behavior*, *Scientific Data*, *Brain Structure and Function*, *Translational Neuroscience*, *Scientific Reports*, *Communications Biology*, *Imaging Neuroscience*, and *Biological Psychiatry*.

PROFESSIONAL MEMBERSHIPS

Organization for Human Brain Mapping

Iranian Medical Council